

V6.2 · Ontology Graph Analysis

The biochemical hierarchy is a graph, not a tree

Objective	Allow a motif to have several parents, and measure the resulting structure.
Method	A directed multi-level graph: motif → L2 chemical theme → L3 biochemical system, with edge weights taken from the soft membership. Centrality and communities are computed on the undirected projection.
Frozen	Atlas 09ed804a40836f4a05a91ba10900cded; the V6 MSS map M is an input, never rewritten.
Reproduce	results/v6_rebuild/code/p10 → p11 → p12 → p13

Part 11 — the biochemical ontology as a GRAPH, not a tree
11 of 17 motifs have more than one parent; red edges are the weaker, shared memberships

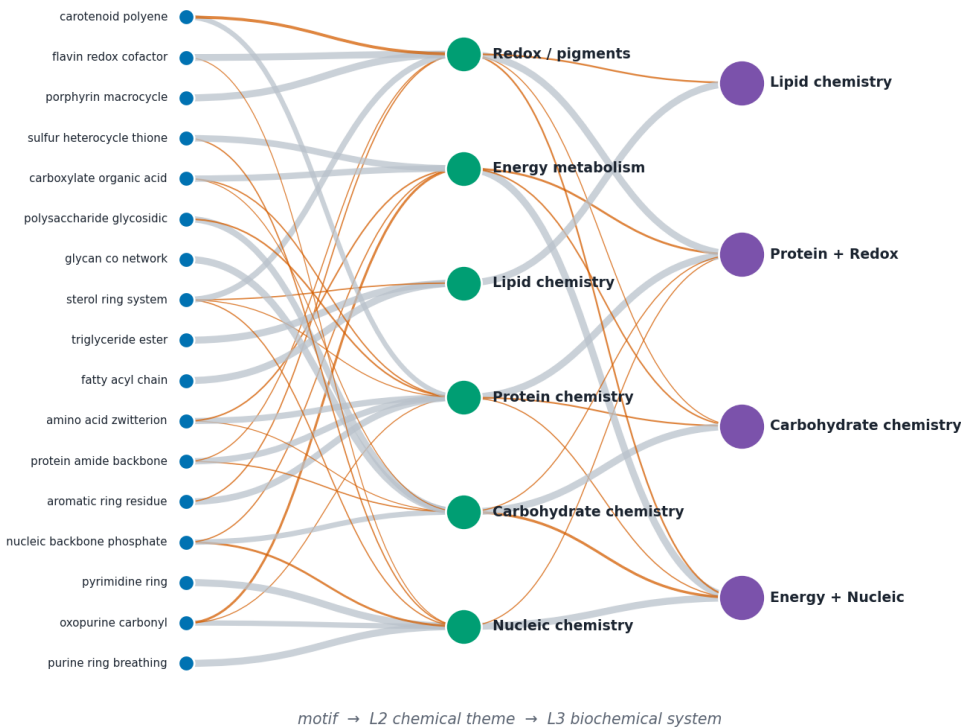


Figure 1 — The ontology graph. Red edges are the weaker, shared memberships.

Results

nodes	25
edges	51
motifs with more than one parent	11 of 17
communities detected	5
multi-parent motifs	oxopurine carbonyl, nucleic backbone phosphate, aromatic ring residue, protein amide backbone, amino acid zwitterion, sterol ring system, polysaccharide glycosidic, carboxylate organic acid, sulfur heterocycle thione, flavin redox cofactor, carotenoid polyene

Highest-betweenness nodes — the bridges between chemistries

node	kind	betweenness
sterol ring system	motif	0.1109
oxopurine carbonyl	motif	0.0181
nucleic backbone phosphate	motif	0.0163
flavin redox cofactor	motif	0.0111
sulfur heterocycle thione	motif	0.0047

Interpretation

11 of 17 motifs have more than one parent, so the tree assumption V6 inherited is simply false for this corpus. The high-betweenness nodes are the chemistries that bridge themes — they are also, unsurprisingly, the motifs with the highest membership entropy and the ones most often confused in the recoverability matrix. Centrality, entropy and confusion are three views of one underlying fact.

An interactive version is written to `figures_v62/v62_ontology_graph.html` (plotly, self-contained, no external data).

Limitations

- L3 edges are derived from the dominant L2 assignment, so the third level is coarser evidence than the second.
- Community detection on 25 nodes is unstable; the communities are reported for description, not inference.
- Edge weights are memberships, not probabilities of chemical presence.

Recommendations

- Use the graph, not the tree, whenever a downstream layer needs a parent — several motifs have no single correct parent.
- Carry betweenness forward as a per-motif transfer-risk score into the Ag-SERS phase.

Atlas 09ed804a40836f4a05a91ba10900cded verified unchanged. All values recomputed from `results/v6_rebuild/`.